Practical - #5 on WAR File using Jetty

```
Command Prompt - mvn jett

C:\Users\narut>mvn archetype:generate   -DgroupId=com.example   -DartifactId=my-webapp   -DarchetypeArtifactId=maven-archetype-webapp  -Darchety
peVersion=1.4   -DinteractiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib
/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.4/maven-archetype-webapp-1.4
.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.4/maven-archetype-webapp-1.4.
jar (6.8 kB at 7.3 kB/s)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Archetype: maven-archetype-webapp:1.4
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: my-webapp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: packageInPathFormat, Value: com/example
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: my-webapp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Project created from Archetype in dir: C:\Users\narut\my-webapp
```

Step 1: Create the template & Get inside the folder

mvn archetype:generate -DgroupId=com.example -DartifactId=my-webapp -DarchetypeArtifactId=maven-archetype-webapp -DarchetypeVersion=1.4 -DinteractiveMode=false

Step 2: Open and edit the pom.xml

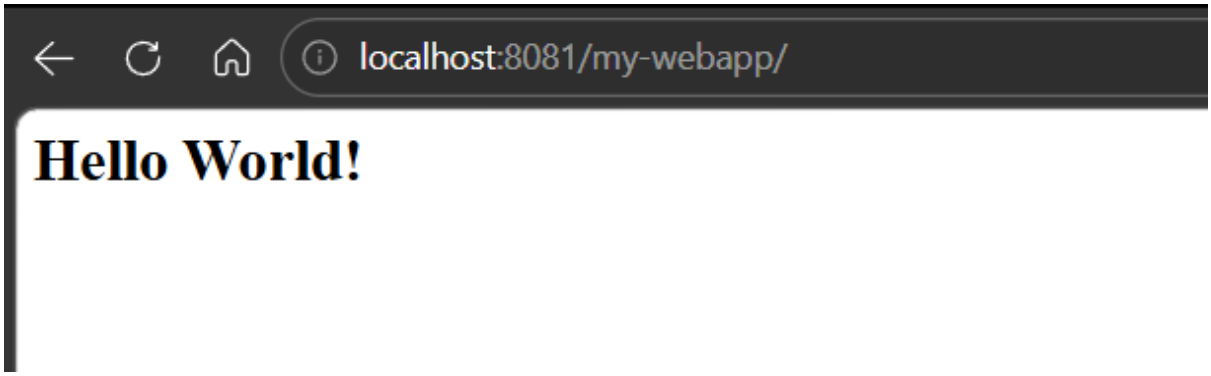
```
C:\Users\narut>cd my-webapp

C:\Users\narut\my-webapp>dir
 Volume in drive C is Windows
 Volume Serial Number is 2461-2C7D

 Directory of C:\Users\narut\my-webapp

31-03-2026  12:35    <DIR>          .
31-03-2026  12:35    <DIR>          ..
31-03-2026  12:35             2,268 pom.xml
31-03-2026  12:35    <DIR>          src
               1 File(s)          2,268 bytes
               3 Dir(s)  10,413,432,832 bytes free

C:\Users\narut\my-webapp>notepad pom.xml
```

Step-3: mvn clean install

```
Command Prompt - mvn jett
C:\Users\narut\my-webapp>mvn clean install
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib
/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:my-webapp >------------------------
[INFO] Building my-webapp 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ war ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.11.0/maven-compiler-plugin-3.11.
0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.11.0/maven-compiler-plugin-3.11.0
.pom (9.8 kB at 22 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.11.0/maven-compiler-plugin-3.11.
0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.11.0/maven-compiler-plugin-3.11.0
.jar (66 kB at 677 kB/s)
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ my-webapp ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ my-webapp ---
[INFO] skip non existing resourceDirectory C:\Users\narut\my-webapp\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ my-webapp ---
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-java/1.1.2/plexus-java-1.1.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-java/1.1.2/plexus-java-1.1.2.pom (5.0 kB at 58 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-languages/1.1.2/plexus-languages-1.1.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-languages/1.1.2/plexus-languages-1.1.2.pom (4.1 kB at 1
01 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.4/asm-9.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.4/asm-9.4.pom (2.4 kB at 25 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.13.0/plexus-compiler-api-2.13.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.13.0/plexus-compiler-api-2.13.0.pom (1.1
 kB at 32 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler/2.13.0/plexus-compiler-2.13.0.pom
```

Step 4: mvn jetty:run

```
C:\Users\narut\my-webapp>mvn jetty:run-war
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib
/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-maven-plugin/9.4.54.v20240208/jetty-maven-plugin-9.4.54.v
20240208.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-maven-plugin/9.4.54.v20240208/jetty-maven-plugin-9.4.54.v2
0240208.pom (12 kB at 19 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-project/9.4.54.v20240208/jetty-project-9.4.54.v20240208.p
om
```

Step-5: Open your browser and visit:

http://localhost:8081